A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.2: Logic Gates and Logic Circuits

Concept 3.2.3: Constructing Logic Circuits

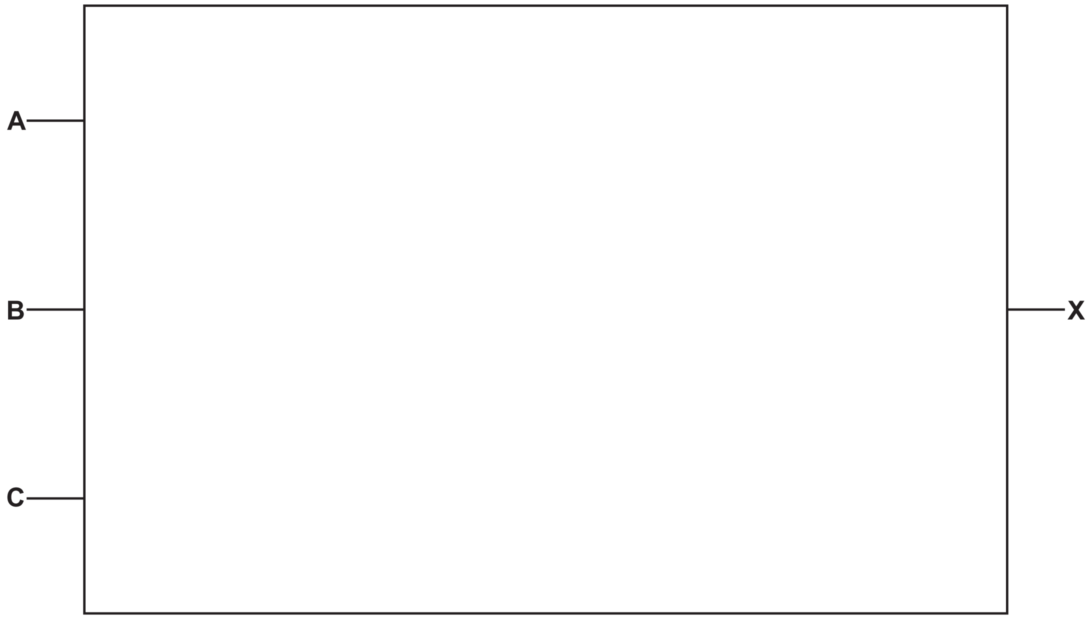
Total Questions: 14

Mark Schemes begin on Page 10

Question 1 | Source: 9618_w25_qp_11 (Q3.a)

3 **(a)** Draw a logic circuit for the logic expression:

X = NOT ((**A** NAND **B**) XOR (NOT **A** OR NOT **C**))

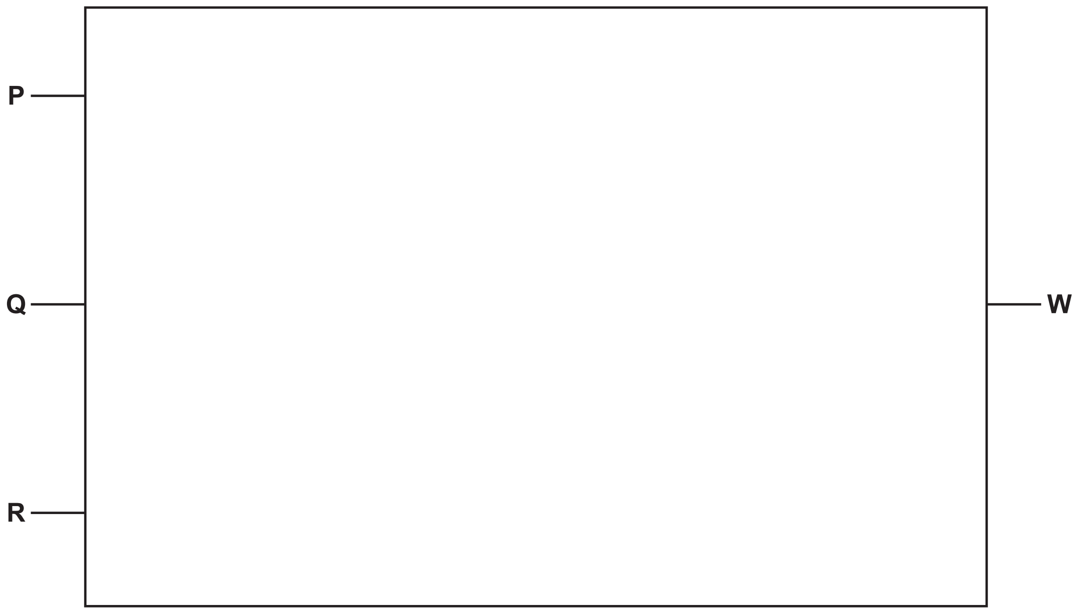


[2]

Question 3 | Source: 9618_w24_qp_12 (Q1.b)

(b) Draw the logic circuit for the logic expression:

W = **P** NAND ((**Q** OR NOT **R**) XOR (**P** XOR **Q**))



[2]

Question 4 | Source: 9618_w24_qp_13 (Q1.c)

(c) Draw the logic circuit for the logic expression:

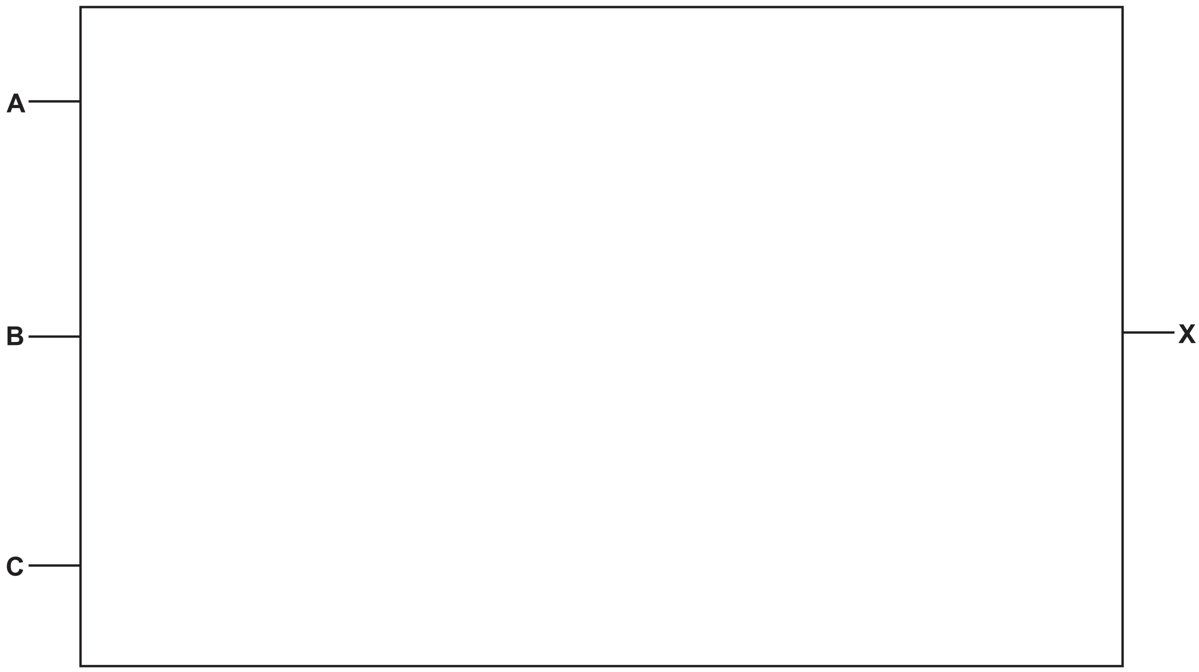
X = ((NOT **A** AND (**B** AND **C**)) OR (**B** NAND **C**)) AND NOT **A**

[2]

Question 5 | Source: 9618_s24_qp_11 (Q1.b)

(b) Draw a logic circuit for the logic expression:

X = NOT (NOT **A** AND (NOT **B** XOR **C**))



[2]

Question 6 | Source: 9618_s24_qp_12 (Q1.b)

(b) Draw a logic circuit for this logic expression:

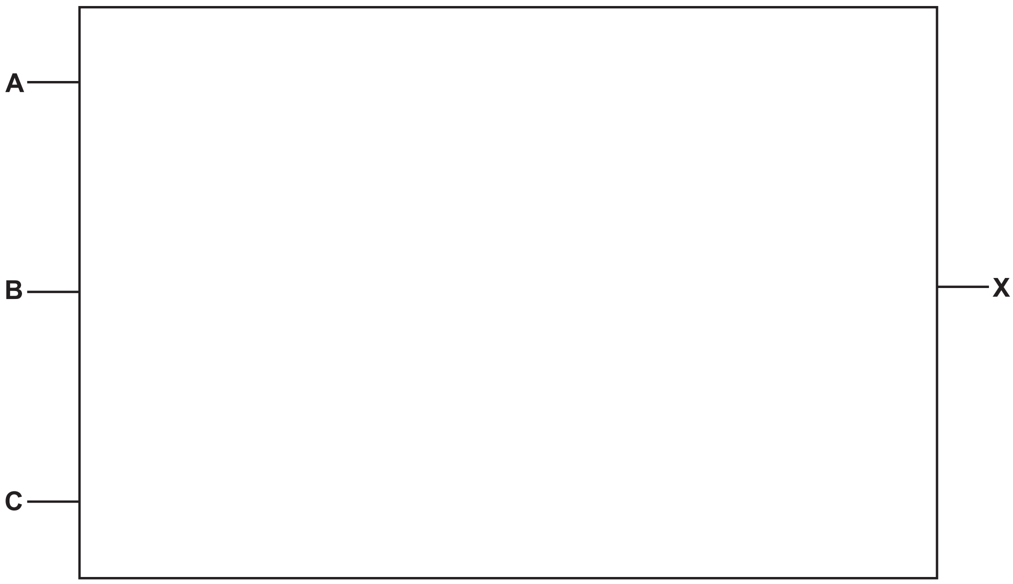
X = NOT ((**A** AND **B**) OR (**C** AND **D**))

[2]

Question 7 | Source: 9618_w23_qp_11 (Q4.b)

(b) Draw a logic circuit for the logic expression:

X = (**A** AND NOT (**B** OR **C**)) AND (**B** NOR **C**)

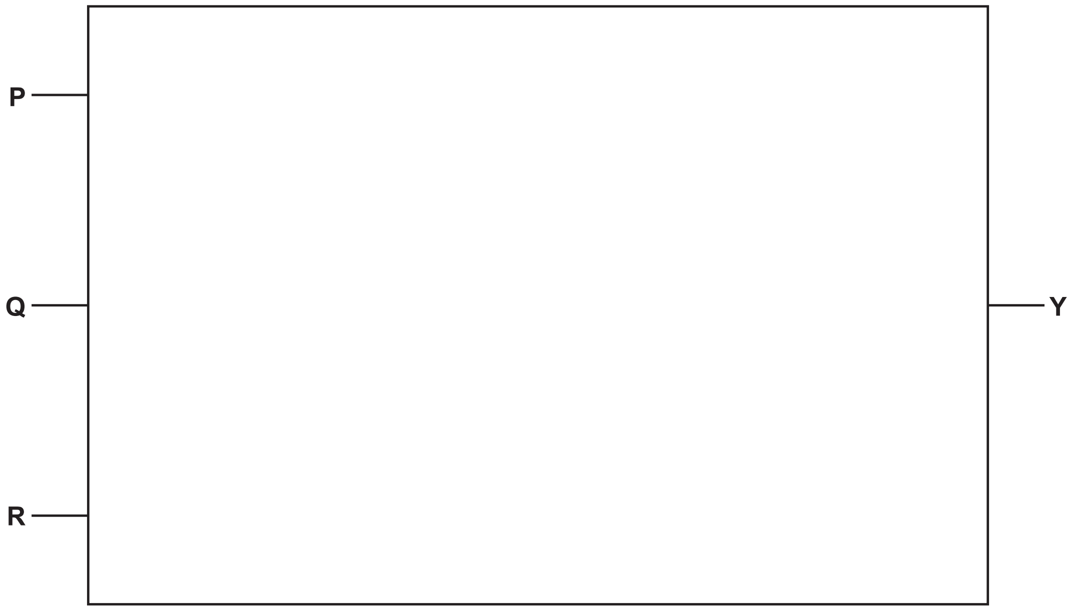


[2]

Question 8 | Source: 9618_w23_qp_13 (Q4.b)

(b) Draw a logic circuit for the logic expression:

Y = ((**P** AND **Q**) XOR ((NOT **Q**) OR **R**)) AND NOT **P**

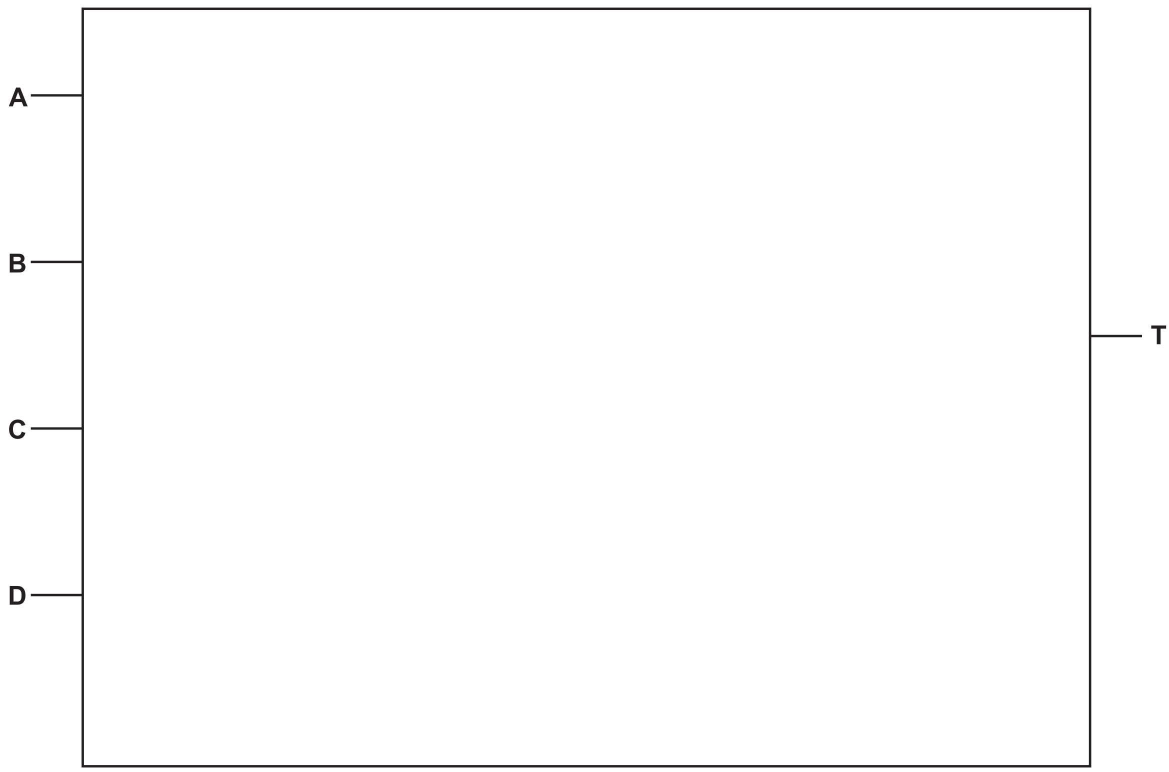


[2]

Question 9 | Source: 9618_s23_qp_11 (Q5.a)

5 **(a)** Draw the logic circuit for this logic expression:

T = (NOT **A** OR **B**) XOR (**C** NAND **D**)

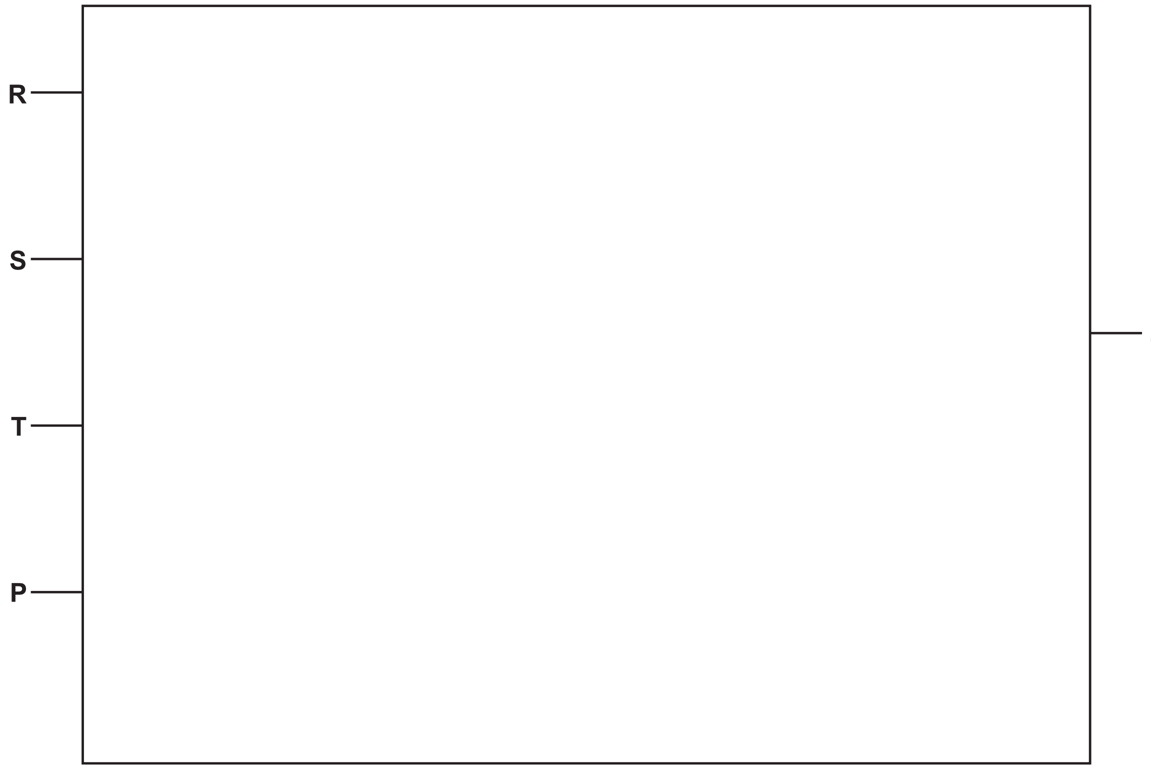


[2]

Question 10 | Source: 9618_s23_qp_12 (Q6.a)

6 **(a)** Draw the logic circuit for this logic expression:

Z = (**R** XOR **S**) AND (NOT **T** NOR **P**)



[2]

Question 11 | Source: 9618_w22_qp_11 (Q3.a)

3 **(a)** Draw a logic circuit for the logic expression:

X = NOT ((NOT **A** AND NOT **B**) OR (NOT **B** AND NOT **C**))

[2]

Question 12 | Source: 9618_w22_qp_12 (Q3.a)

3 **(a)** A greenhouse has an automatic window.

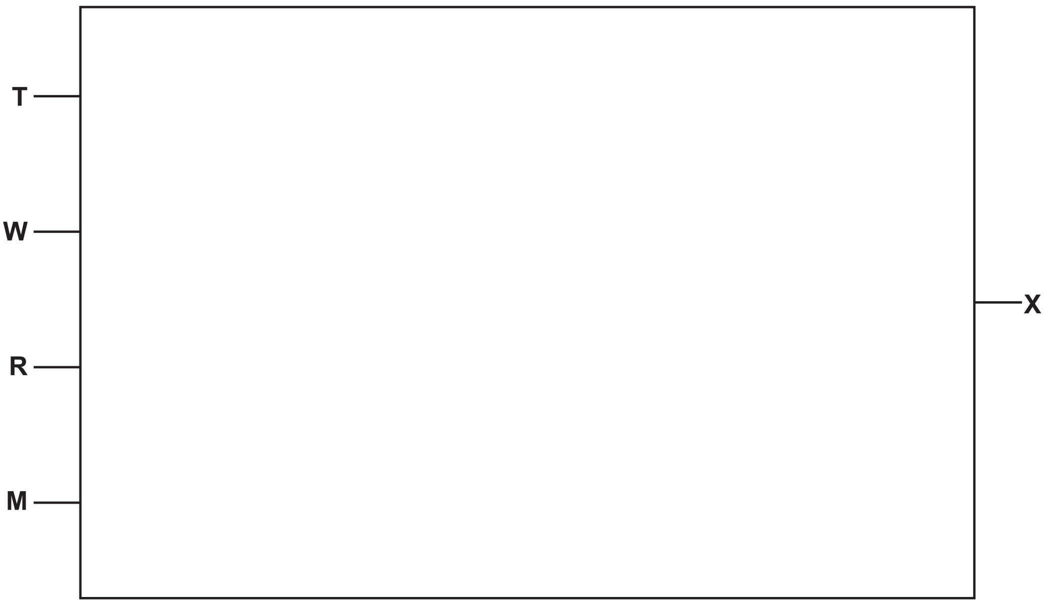
The window (**X**) operates according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
T	Temperature	1	Too high
		0	Acceptable
W	Wind speed	1	Too high
		0	Acceptable
R	Rain	1	Detected
		0	Not detected
M	Manual override	1	On
		0	Off

The window opens (**X** = 1) if:

- the temperature is too high **and** the wind speed is acceptable
- **and**
- rain is not detected, **or** the manual override is off.

Draw a logic circuit to represent the operation of the window.

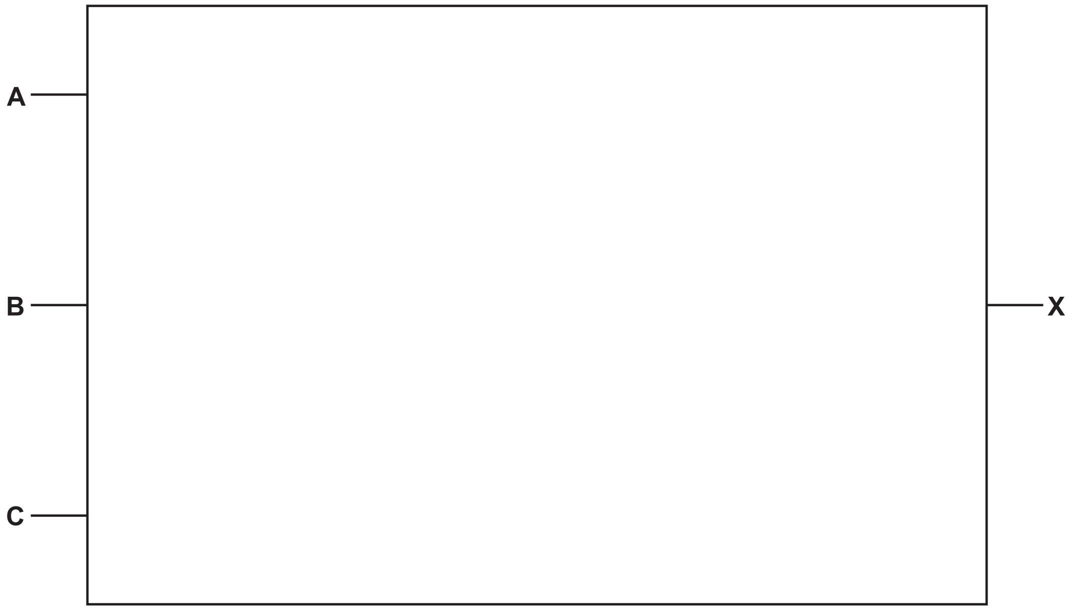


[3]

Question 13 | Source: 9618_w22_qp_13 (Q5.a)

5 **(a)** Draw a logic circuit for the logic expression:

X = NOT ((NOT (**A** AND **B**)) OR (NOT (**B** AND **C**)))



[3]

Question 14 | Source: 9618_w21_qp_12 (Q2.b)

(b) Draw a logic circuit for the following logic expression.

X = NOT(NOT(**A** AND **B**)AND **C**)

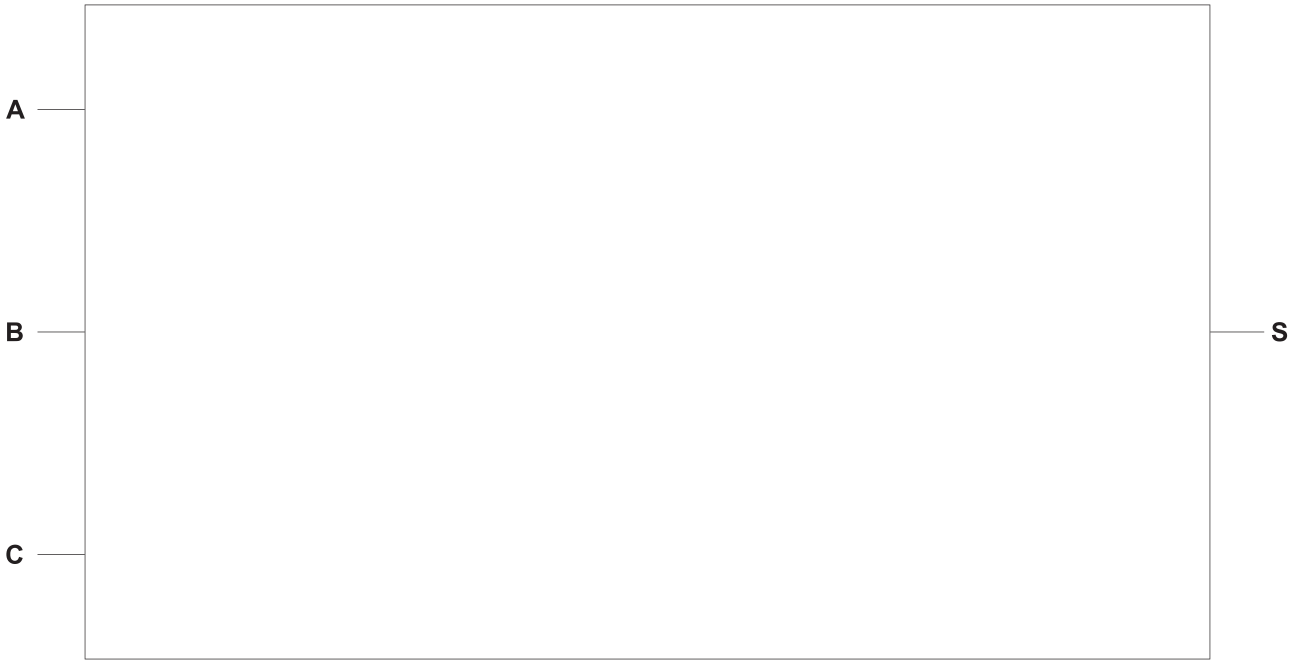
[2]

Question 15 | Source: 9618_s21_qp_12 (Q3.a)

3 A logic expression is given:

S = (**A** AND **B** AND **C**) OR (**B** XOR **C**)

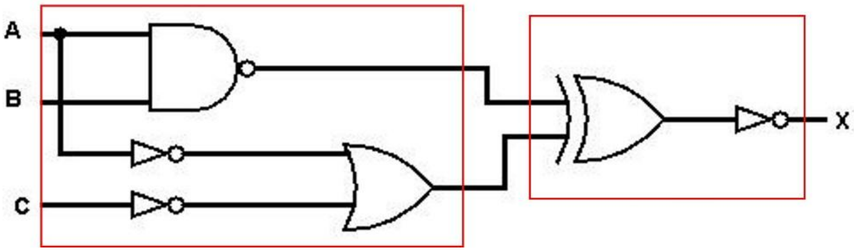
(a) Draw the logic circuit for the given expression.



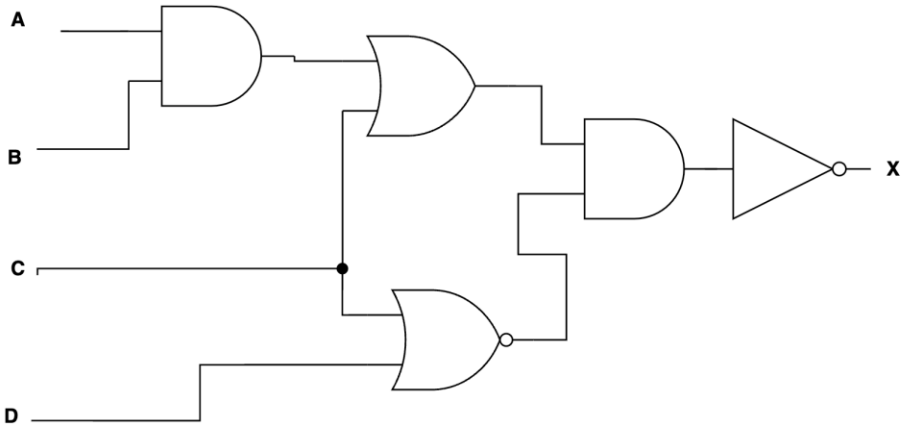
[4]

MARK SCHEMES

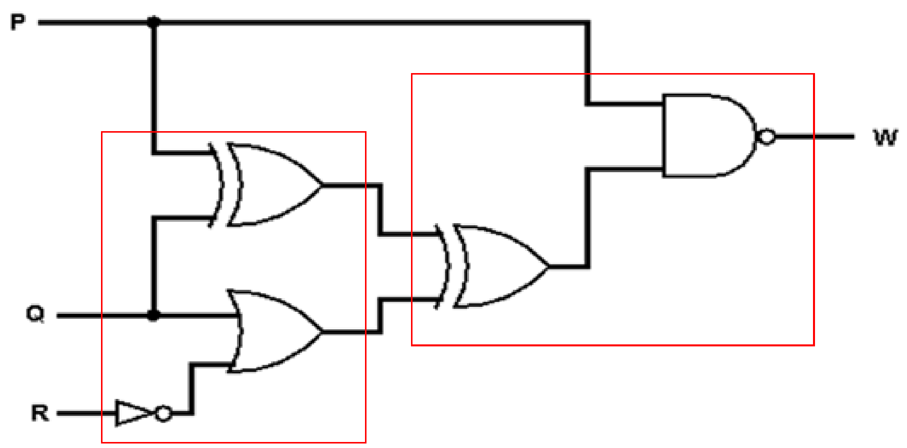
Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q3.a)

3(a)	**One mark** per bullet point, **max 2** marks • (A NAND B) **and** (NOT A OR NOT C) • XOR and NOT  	**2**

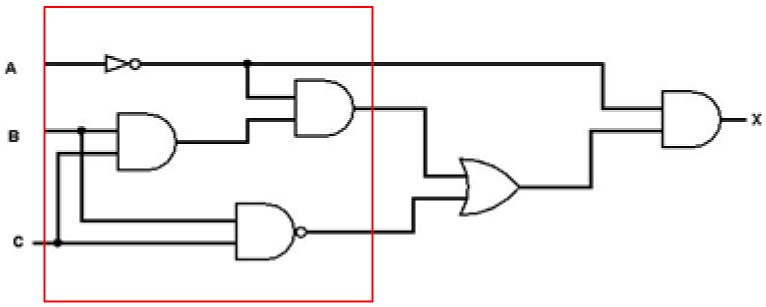
Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q3.b)

3(b)	**1 mark** per bullet point, **max 2** marks • A AND B OR C • C NOR D and final two gates  	**2**

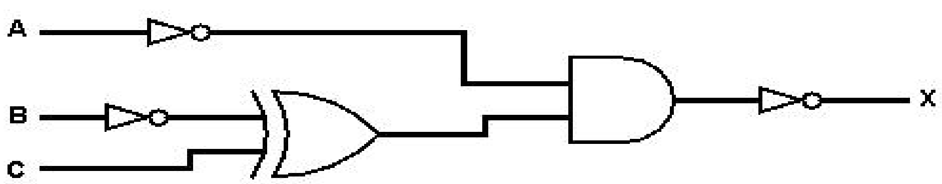
Mark Scheme for Question 3 | Source: 9618_w24_ms_12 (Q1.b)

1(b)	**1 mark** for (P XOR Q) and (Q OR NOT R) **1 mark** for second XOR gate and final NAND gate with appropriate inputs 	**2**

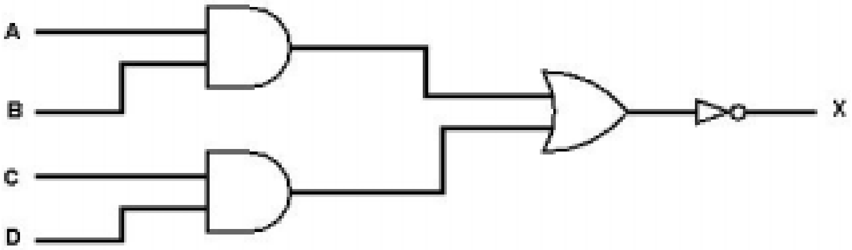
Mark Scheme for Question 4 | Source: 9618_w24_ms_13 (Q1.c)

1(c)	**1 mark** for NOT gate, 2 correct AND gates and the NAND gate **1 mark** for OR gate and the final AND gate 	**2**

Mark Scheme for Question 5 | Source: 9618_s24_ms_11 (Q1.b)

1(b)	**1 mark** for NOT B XOR C **1 mark** for NOT A and final AND plus NOT 	**2**

Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q1.b)

<table>
<tr>
<td>1(b)</td>
<td>1 mark for both AND gates with correct inputs
1 mark for correct OR and NOT gates with correct inputs and no superfluous gates:

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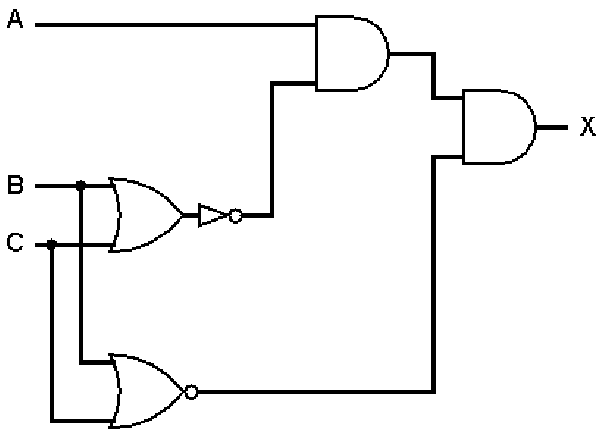
</table>

Mark Scheme for Question 7 | Source: 9618_w23_ms_11 (Q4.b)

<table>
<tr>
<td>4(b)</td>
<td>1 mark for both AND gates
1 mark for NOT gate <u>and</u> OR gate <u>and</u> NOR gate

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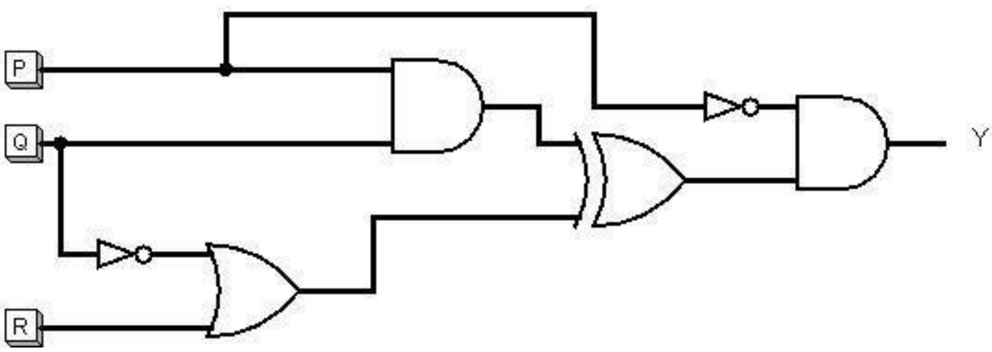
Mark Scheme for Question 8 | Source: 9618_w23_ms_13 (Q4.b)

<table>
<tr>
<td>4(b)</td>
<td>1 mark for P AND Q <u>and</u> NOT Q OR R
1 mark for NOT gate and AND gate and XOR gate

</td>
<td>2</td>
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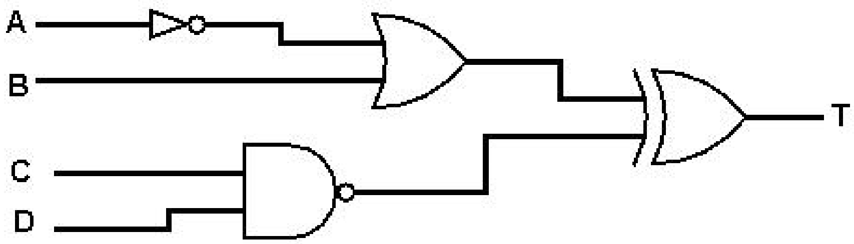
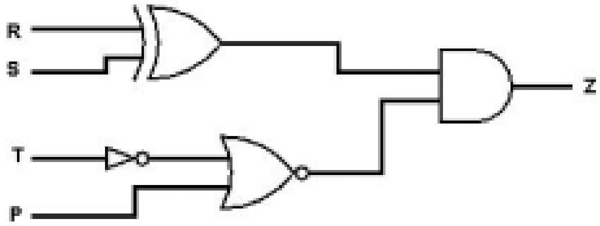
</table>

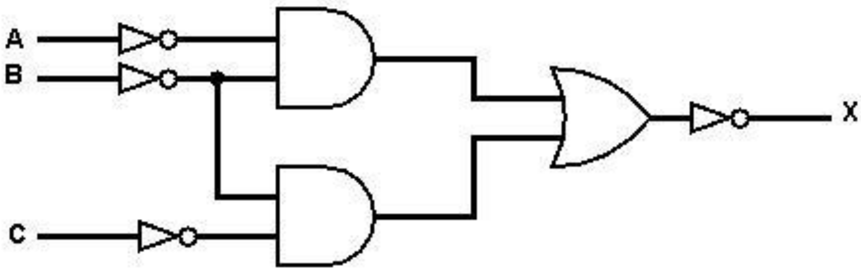
Mark Scheme for Question 9 | Source: 9618_s23_ms_11 (Q5.a)

5(a)	**1 mark** for 2 gates **2 marks** for all 4 gates 	**2**

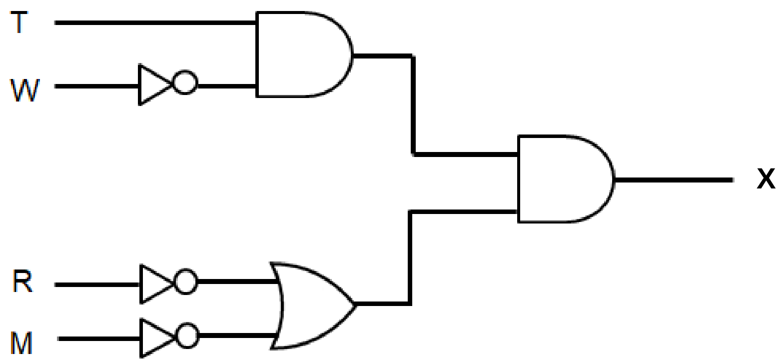
Mark Scheme for Question 10 | Source: 9618_s23_ms_12 (Q6.a)

6(a)	**1 mark** for correct XOR and AND gates, **with** correct inputs **1 mark** for correct NOT and NOR gates **with** correct inputs 	**2**

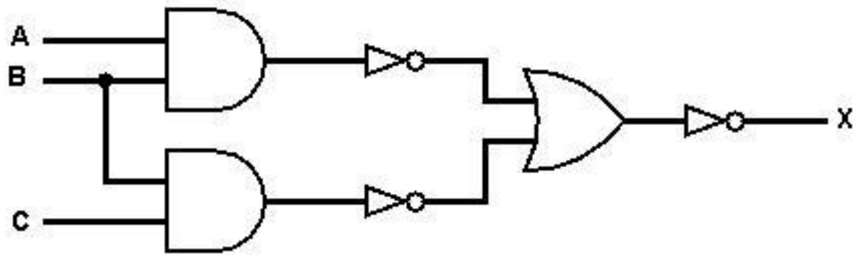
Mark Scheme for Question 11 | Source: 9618_w22_ms_11 (Q3.a)

3(a)	**1 mark** for each bullet point: • NOT A AND NOT B **and** NOT B AND NOT C // A NOR B **and** B NOR C • final OR and NOT gates (with correct inputs) // NOR gate (with correct inputs) 	**2**

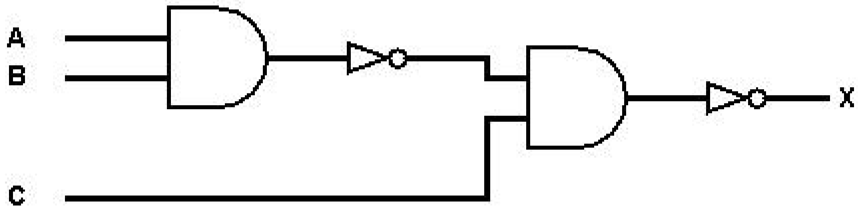
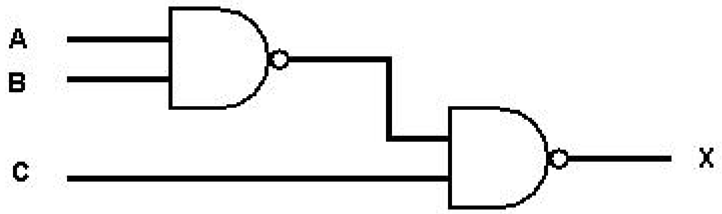
Mark Scheme for Question 12 | Source: 9618_w22_ms_12 (Q3.a)

3(a)	**1 mark** for T AND NOT W **1 mark** for NOT R OR NOT M **1 mark** for final AND 	**3**

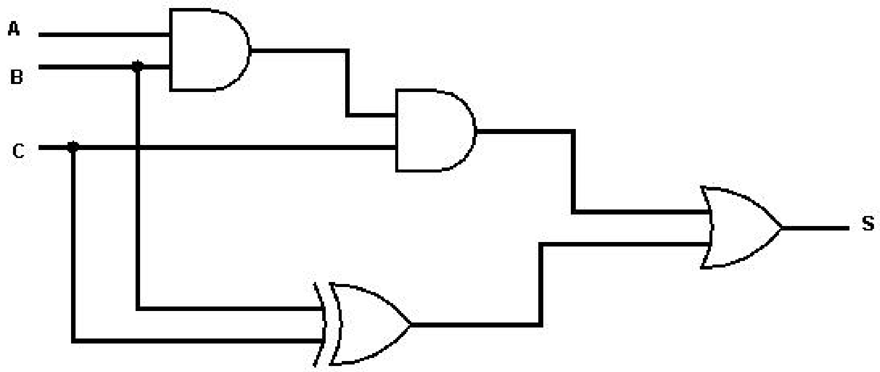
Mark Scheme for Question 13 | Source: 9618_w22_ms_13 (Q5.a)

5(a)	**1 mark** for each bullet point: • NOT (A AND B) • NOT (B AND C) • NOT(NOT(A AND B) OR NOT(B AND C)) 	**3**

Mark Scheme for Question 14 | Source: 9618_w21_ms_12 (Q2.b)

2(b)	**1 mark** for each correct bullet point • NOT (A AND B) // A NAND B • NOT the result AND C // the result NAND C  OR  	**2**

Mark Scheme for Question 15 | Source: 9618_s21_ms_12 (Q3.a)

3(a)	**1 mark** for each correct gate, with correct inputs  	**4**